MASTER CLASS DELIVERABLE • VIDEO #12

Real-Time AI Voice Agent Architecture: LiveKit + WebRTC

Sub-300ms Conversational Voice Agents with Natural Turn-Taking & Tool Execution

FRAMEWORK: LIVEKIT AGENTS (PYTHON) STT/TTS: DEEPGRAM NOVA-2 / CARTESIA / GEMINI LIVE PUBLISHED BY: @PARKER.RENDER

1. The 300ms Glass-Smooth Voice Architecture

Traditional voice pipelines chain HTTP requests (Whisper STT -> GPT-4 Text -> ElevenLabs TTS), resulting in frustrating 2.5s+ latency. **LiveKit Agents** uses bidirectional WebRTC audio streaming to process audio chunks in real time with Voice Activity Detection (VAD) and instant interruption cancellation.

2. Installation & Prerequisites

```
pip install livekit-agents livekit-plugins-deepgram livekit-plugins-openai livekit-plugins-silero
```

3. Production Python Voice Agent (agent.py)

```
import asyncio
from livekit.agents import AutoSubscribe, JobContext, WorkerOptions, cli, llm
from livekit.agents.voice_assistant import VoiceAssistant
from livekit.plugins import deepgram, openai, silero

async def entrypoint(ctx: JobContext):
    await ctx.connect(auto_subscribe=AutoSubscribe.AUDIO_ONLY)

    chat_context = llm.ChatContext().append(
        role="system",
        text="You are a professional voice receptionist for Parker Studio. Speak naturally in biling
    )

    assistant = VoiceAssistant(
        vad=silero.VAD.load(),
        stt=deepgram.STT(),
        llm=openai.LLM(model="gpt-4o-mini"),
        tts=openai.TTS(voice="alloy"),
        chat_ctx=chat_context,
    )

    assistant.start(ctx.room)
    await assistant.say("Namaste! Parker Studio me aapka swagat hai. Main aapki kya madad kar sakta

if __name__ == "__main__":
    cli.run_app(WorkerOptions(entrypoint_fnc=entrypoint))
```

4. Key Production Features

- **Natural Interruptions:** User bolna shuru karte hi assistant bina latency ke chup ho jaata hai.

- **Function Calling:** Direct Google Calendar API call karke appointments auto-schedule karna.
- **Telephony SIP Trunking:** Normal phone numbers (Twilio / Plivo) se direct WebRTC bridge.

Crafted by **@parker.render** • Visit parker-render.netlify.app/call for updates.